



microring modulators.



contents

- Charge solver.
- PN junction simulation.
- Plan



contents



Charge solver.



➤ Charge Solver



- The Charge Solver is a physics-based electrical simulation tool for semiconductor devices, which solves the system of equations describing the electrostatic potential (Poisson's equation) and density of free carriers (the drift-diffusion equations), producing accurate results for a wide range of semiconductor devices under common operating conditions.



contents

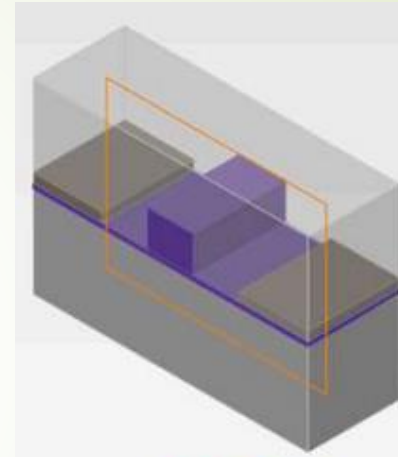


PN junction simulation.



PN junction simulation.

- The effect of bias voltage on carriers' concentration.
- Plot capacitance vs Voltage.

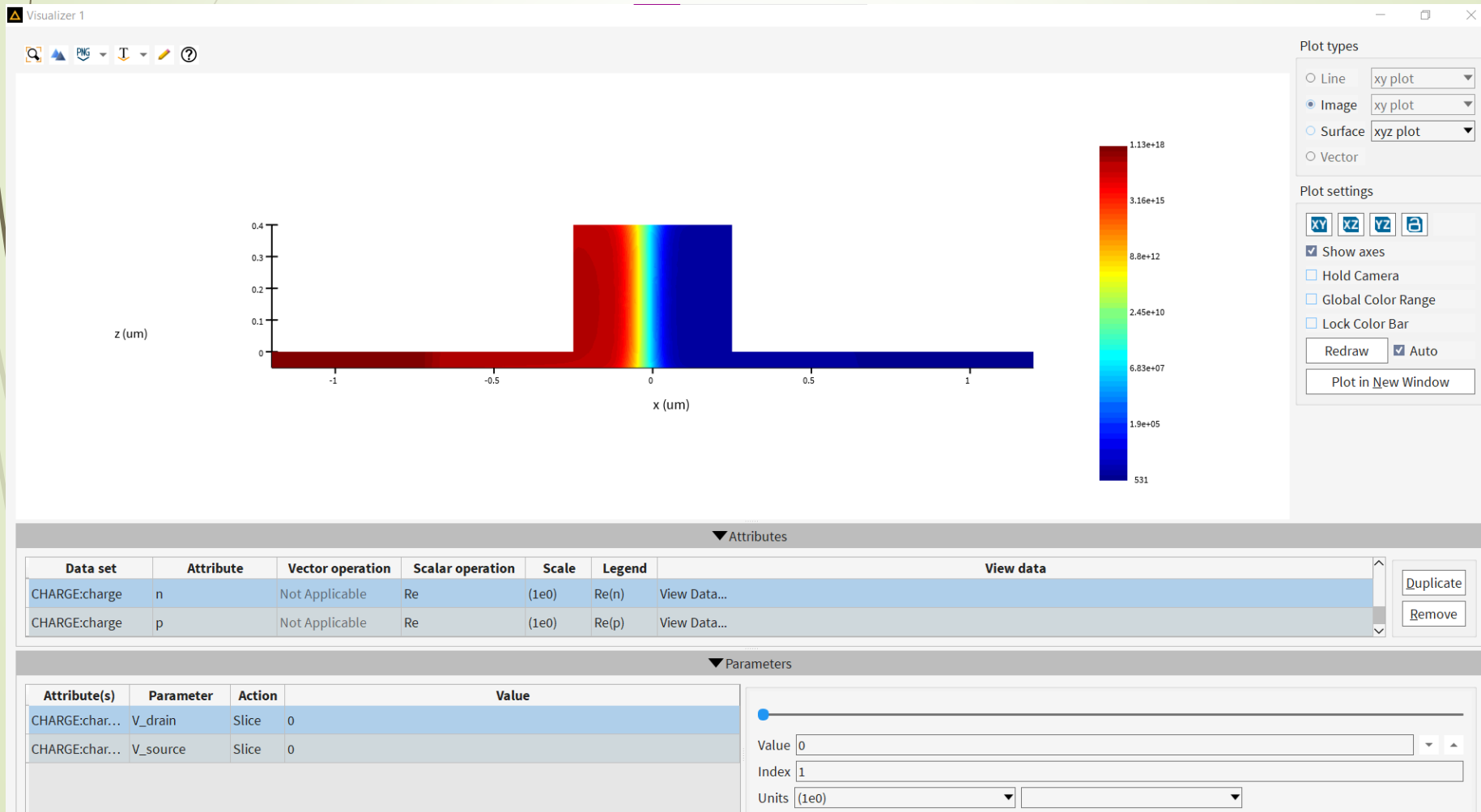


$$W_d \approx \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} - V_a)}$$

$$(c_j = 1/W_d)$$

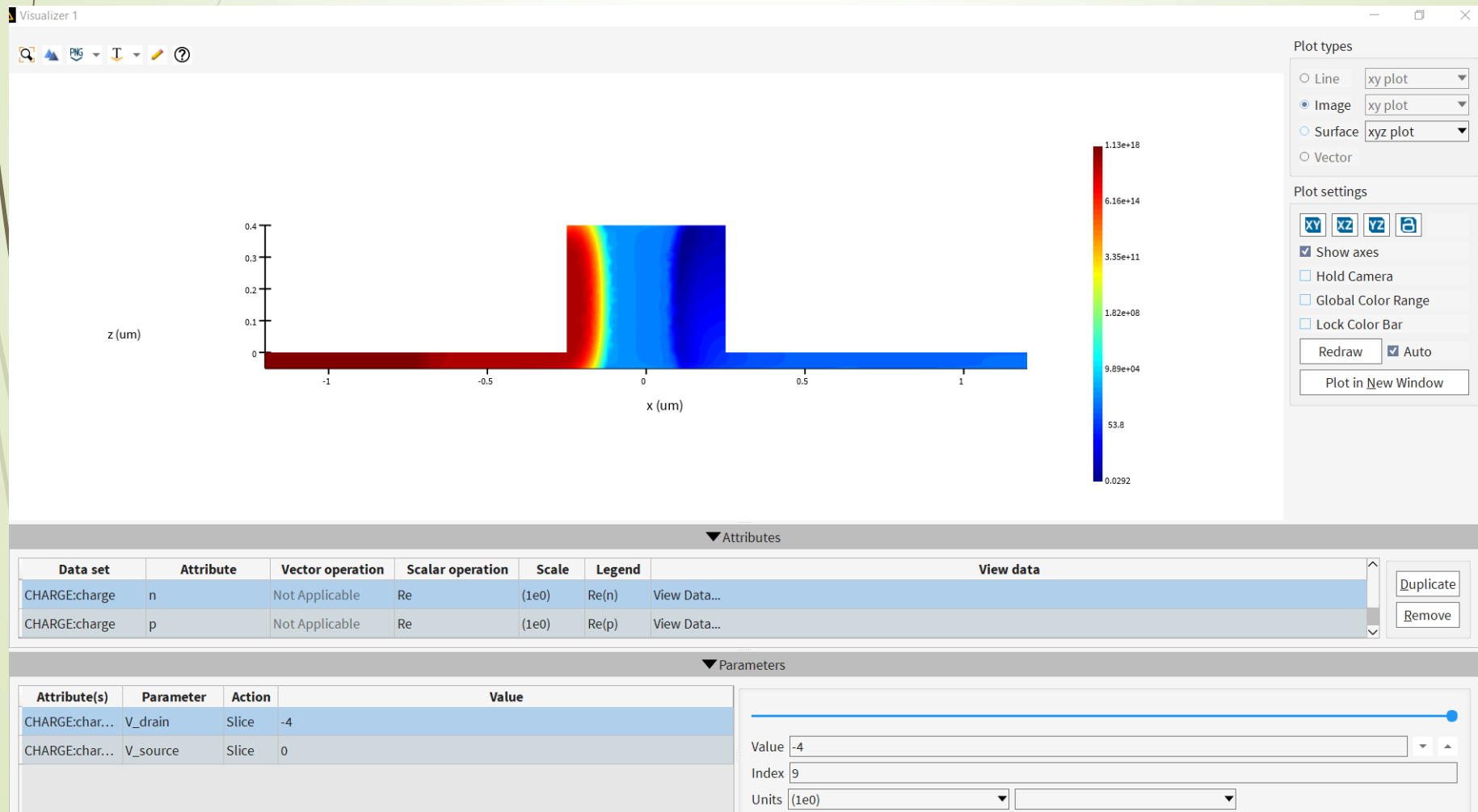
PN junction simulation.

- Charge Distribution Profile (log scale)



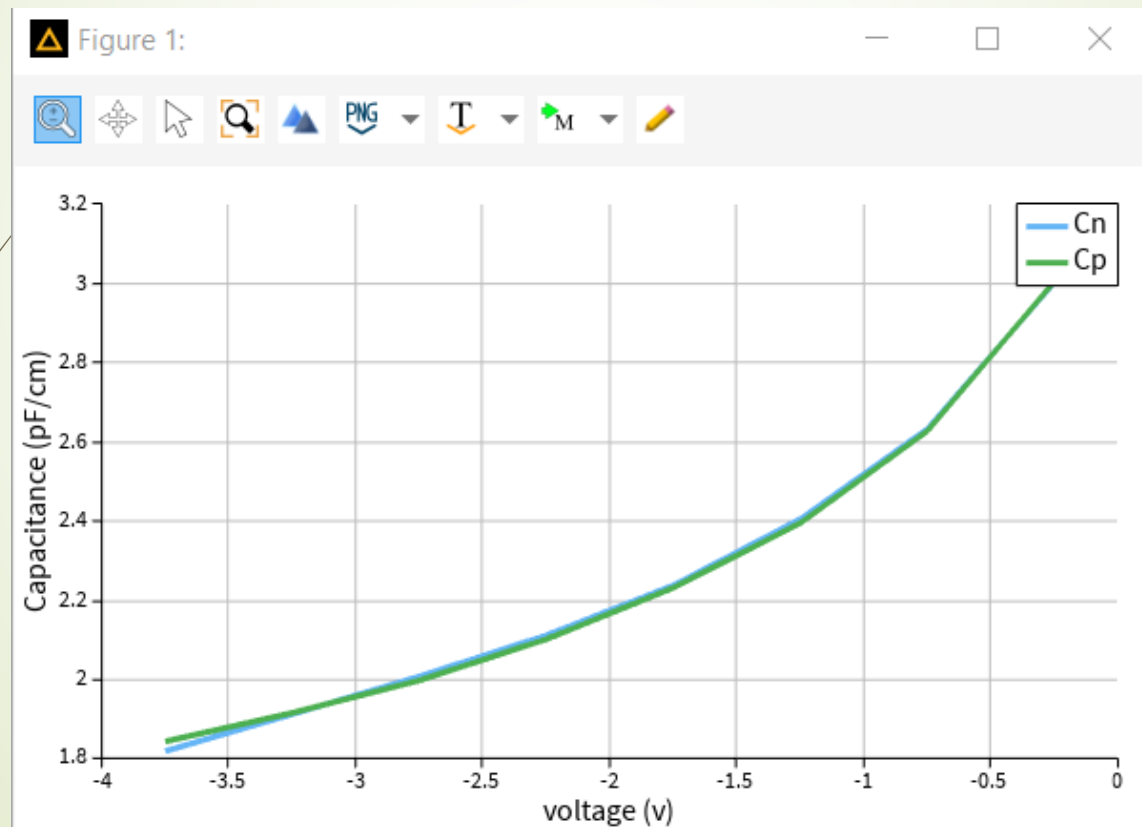
PN junction simulation.

- Charge Distribution Profile (log scale)



PN junction simulation.

- C-V plot of the device





contents



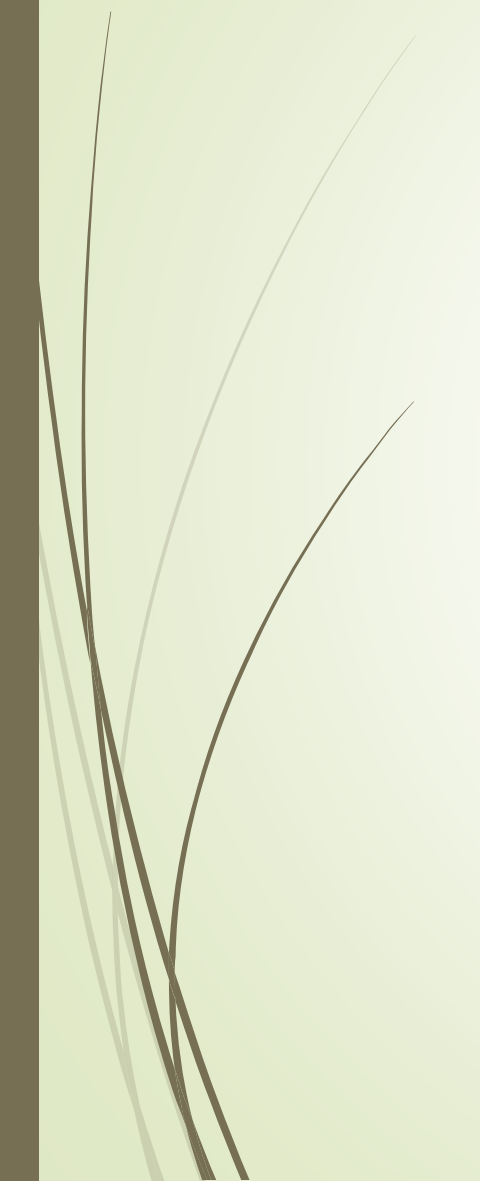
Plan





Plan

- Device simulations.
- Finalize modulator design and geometry.
- Determine performance metrics.
- Performance review.
- Documentation.



Thanks 😊